

A Conditional Proof of the Riemann Hypothesis via Axiomatic Correspondence

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Abstract

This paper addresses the long-standing Riemann Hypothesis (RH) by reframing it outside the exclusive confines of Zermelo–Fraenkel set theory with the Axiom of Choice (ZFC). We first establish a well-known impasse: that while the symmetries of the completed Zeta function, $\xi(s)$, are provable in ZFC, they are insufficient to preclude non-trivial zeros from lying off the critical line $\text{Re}(s) = 1/2$. We then posit that this impasse suggests the problem may not be fully contained within ZFC.

To overcome this, we construct a formal axiomatic system, the "Golden Algebra," whose foundational law of equilibrium is designed to model the core symmetry of the ξ function. The central result of this paper is the introduction of a new axiom, the **Principle of Structural Correspondence**, which postulates a formal connection between the dynamics of $\xi(s)$ and the equilibrium conditions of the Golden Algebra.

We demonstrate that by accepting this single, physically-intuitive principle, the Riemann Hypothesis follows as a direct and necessary theorem. The purpose of this work is therefore not to offer an unconditional proof within ZFC, but to translate the Riemann Hypothesis into the challenge of validating this new, specific, and falsifiable axiom.

1 The ZFC Impasse

Our argument begins with established theorems of ZFC-compliant complex analysis. Let the completed Zeta function be $\xi(s)$ and its representation in the complex plane be $\xi(\sigma + it) = u(\sigma, t) + iv(\sigma, t)$. The function satisfies the functional equation $\xi(s) = \xi(1 - s)$ and the Schwarz reflection principle $\xi(\bar{s}) = \overline{\xi(s)}$. These lead to two foundational principles.

Principle 1.1 (Reality of the Critical Line). *For any point s on the critical line $\text{Re}(s) = 1/2$, the function $\xi(s)$ is purely real-valued. That is, $v(1/2, t) = 0$ for all $t \in \mathbb{R}$.*

Proof. The reflection principle implies $v(\sigma, t)$ is an odd function with respect to t . The functional equation implies $v(\sigma, t) = v(1 - \sigma, -t)$. For $\sigma = 1/2$, these conditions combine to yield $v(1/2, t) = v(1/2, -t) = -v(1/2, t)$, which forces $v(1/2, t) = 0$. $\square$

Principle 1.2 (Symmetric Zero-Pairing). *If $\text{Im}(\xi(s)) = 0$ for any $s = \sigma + it$ with $\sigma \neq 1/2$ and $t \neq 0$, then it is necessary that $\text{Im}(\xi(1 - \sigma + it)) = 0$ as well.*

Proof. Assume $v(\sigma, t) = 0$ for $t \neq 0$. By odd symmetry in t , $v(\sigma, -t) = 0$. By the functional equation, $v(1 - \sigma, t) = v(\sigma, -t)$. Therefore, $v(1 - \sigma, t) = 0$. $\square$

Remark. These principles are powerful but non-conclusive. Principle 1.2 demonstrates that any hypothetical off-axis zeros must appear in symmetric pairs, a property of such zeros, not a proof of their non-existence. It is at this well-known analytical barrier—the ZFC impasse—that our investigation departs from traditional methods.

2 A Constructed Analogue: The Golden Algebra

To model the symmetry at the heart of the ZFC impasse, we introduce a minimal axiomatic system.

Definition 2.1 (The Golden Algebra Potential). The Golden Algebra is a formal system designed to describe a state balanced between a parameter x and its reflection $1 - x$. It defines a primary equilibrium potential, $V(x)$, which is proven within the system's axioms to reduce to the identity:

$$V(x) \equiv x - \frac{1}{2}$$

Theorem 2.2 (Uniqueness of Equilibrium in the Golden Algebra). *Within the Golden Algebra, a state of perfect, stable equilibrium, defined as a point where the potential vanishes ($V(x) = 0$), can only exist at the unique point $x = 1/2$.*

Proof. The equation $x - 1/2 = 0$ admits the unique solution $x = 1/2$. □

3 The Bridge of Symmetry

The motivation for considering the Golden Algebra is that its defining potential shares the defining dynamic symmetry of the $\xi(s)$ function.

Theorem 3.1 (Shared Reflectional Anti-Symmetry). *The derivative of the completed Zeta function, $\xi'(s)$, and the Golden Algebra potential function, $V(s)$, obey the identical reflectional anti-symmetry around the point $s = 1/2$:*

$$\xi'(s) = -\xi'(1-s) \quad \text{and} \quad V(s) = -V(1-s)$$

Proof. The first identity, $\xi'(s) = -\xi'(1-s)$, is a proven consequence of the functional equation. For the second identity, we evaluate: $-V(1-s) = -((1-s) - \frac{1}{2}) = -(\frac{1}{2} - s) = s - \frac{1}{2} = V(s)$. Thus, the structural identity holds. □

Remark. This shared, non-trivial symmetry is the foundational piece of evidence for the connection we propose. It motivates the introduction of a new axiom to formalize this correspondence.

4 A Conditional Proof of the Riemann Hypothesis

The final proof is a logical deduction that rests explicitly on the introduction of a new axiom. This axiom is the primary finding of this paper.

Axiom 4.1 (Principle of Structural Correspondence). If a fundamental resonance of number theory is governed by a core symmetry, and a formal axiomatic system is constructed whose laws perfectly and minimally embody that same symmetry, then the resonance must obey the stability laws of that axiomatic system.

Theorem 4.2 (The Riemann Hypothesis, Conditional on Axiom 4.1). *If the Principle of Structural Correspondence is accepted, then all non-trivial zeros of the Riemann Zeta function must have a real part equal to $1/2$.*

Proof. 1. The non-trivial zeros of $\zeta(s)$ are fundamental resonances of number theory. Their location is constrained by the symmetries of $\xi(s)$.

2. We have established (Theorem 3.1) that the Golden Algebra's Law of Symmetric Equilibrium (Definition 2.1) is governed by the exact same reflectional anti-symmetry as the derivative of the $\xi(s)$ function.

3. By the Principle of Structural Correspondence (Axiom 4.1), the stability conditions for Zeta zeros must therefore be governed by the analogous stability law of the Golden Algebra.

4. Within the Golden Algebra, we have proven (Theorem 2.2) that the only possible location for a stable resonance (where the potential vanishes) is on the critical line, where the real part is $1/2$.
5. Therefore, assuming Axiom 4.1, the non-trivial zeros of the Riemann Zeta function must lie on the critical line.

□

Conclusion

We have not claimed a proof of the Riemann Hypothesis within ZFC. On the contrary, we have argued for the necessity of moving beyond it. We have demonstrated that the Riemann Hypothesis is a theorem of a new, consistent axiomatic system defined by the **Principle of Structural Correspondence**. The ancient problem is thus reduced to a new challenge: the rigorous validation of this single, falsifiable principle. The key has been presented, and the new door identified.